

Deep Generative Models

Chapter 7: Multimodal Models and Conditional Generation

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Multimodal Model

- + We studied a lot, but did not hear too much about *multimodal models*!
Where do we learn about them?
- Well! You already know *whatever you need to know* about them

Multimodal Model

Multimodal models combine *inputs* from *different modalities*, e.g., visual and textual, to make *outputs* that are *more informed* than *unimodal* models

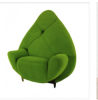
In practice, we always think about *multimodal generative models*

Multimodal Generative Model

A multimodal *generative* model learns to *sample data of one modality* upon observing *samples of data in another modality*

Multimodal Model: *Example*

There are various forms of **multimodal generation**: *text-to-image* conversion

Text Prompt	an armchair in the shape of an avocado. . .				
AI Generated images					

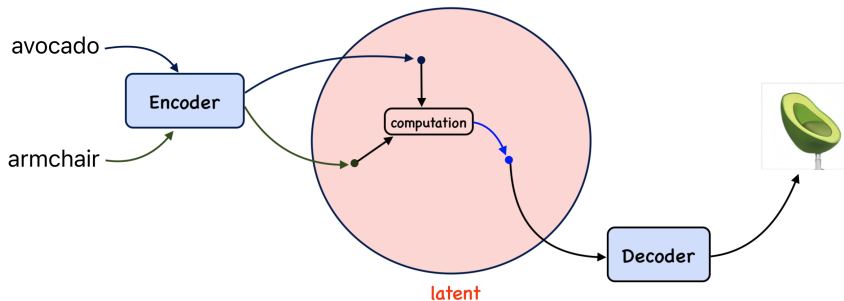
or *video* question *answering*

 Model: No	 Model: Yes
Prompt	Is Emily walking with her bike?

Modality: Latent Space

- + *How do these models work in practice?*
- Basically, by connecting modalities in the **latent space**

Different **modalities** can be related in the **same latent space**



Statistical Viewpoint

- + But, how can we integrate **multimodality** into the **statistical framework**?
- It's all about **conditioning**

Say we see a **sample** of some modality: let's call it u

↳ For instance, u could be a **text describing an image** we are looking for

To change the **modality**, we should **sample x of different nature**

↳ In our example, x is the **image described by u**

The sample x should **depend on u** , so we know that

It's not simply coming from **distribution $P(x)$** , but it's coming from

$$P(x|u)$$

which describe the **density** of samples related to u

Statistical Viewpoint: Conditional Generation

Conditional Generative Model

A **conditional generative model** is a model that upon **being prompted by** u samples from the **conditional** data distribution $P(x|u)$

Example: Say we are converting **text** to **image**

$u =$ give me an image of a dog with glasses

$P(x|u)$ then describes the distribution of all **dogs** with **glasses**



large $P(x|u)$

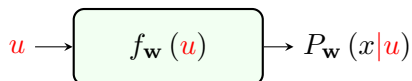


$P(x|u) \approx 0$

Learning Conditional Distribution

- + Can we use our approaches to learn **conditional distributions**?
- Absolutely!

We are now aiming to learn $P(x|u)$: we consider a **computational model**



This model describes the **conditional distribution** for an **input prompt** u

- ↳ This can be **any of models we had**
- ↳ We now also **include the conditioning** in our computation

Learning Conditional Distribution

- + *What about data samples?*
- We collect samples from the **conditional distribution**

$u =$ give me an image of a dog with glasses



- + Do we have **many samples** for any u ?
- We would be fine with **as small as one!**

Learning Joint Distribution

- + With only one pair (u, x) how do we know that the model is **not** learning the **joint distribution**?
- Let it learn, **it's still OK!**

Bayes' rule indicates that

$$P(x|u) = \frac{P(x, u)}{P(u)}$$

For a **fixed** u : $P(x|u)$ is a **normalized** version of $P(x, u)$

- ↳ $P(x^1, u)$ and $P(x^2, u)$ are **relatively** the same as $P(x^1|u)$ and $P(x^2|u)$
- ↳ Learning $P(x, u)$ still does the job for us!

Naive Approach: Label Sampling

- + How can we learn the **conditional** distribution **in practice** then?
- Most naively, we can train our model on **down-sampled data**

Say we want to train sample on **MNIST** with u being the **digit label**

- We can make a subset of MNIST for each **digit** u

$$\mathbb{D}_u = \{x \in \mathbb{D} : \text{label}(x) = u\}$$

- Training any generative model on $\mathbb{D}_u$ learns $P(x|u)$
- + You mean train **10 separate models?!?**
- **Theoretically** yes, but **practically** not really!

Practical Implementation: Label Embedding

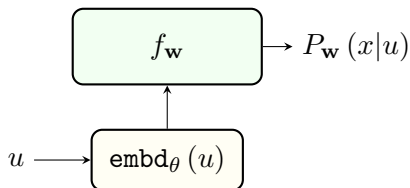
In practice, we can *embed* the label into our model

Let $\text{embd}_\theta(\cdot)$ be an embedding model returning $\text{embd}_\theta(u)$ for a given u

↳ We did this in LMs to *embed a token*

↳ We did this in DPMs to *embed the time*

We can modify our generative model $P_{\mathbf{w}}$ as



and use both MNIST *images* and *their labels* to train this model

Generic Solution: Condition Embedding

*In practice, we always condition a generative model by **embedding***

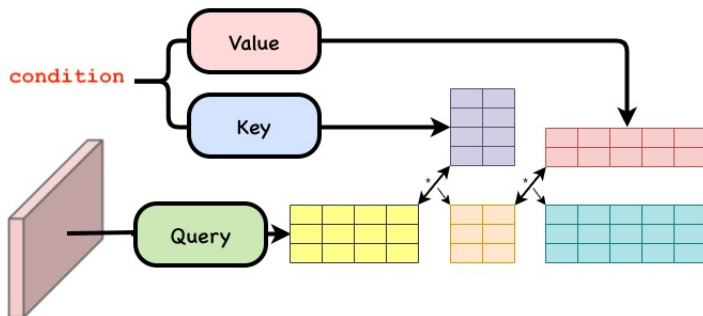
Condition Embedding

*In **condition embedding**, we compute a **rich representation of condition u** via a **computational sub-model**, e.g., an embedding layer or an encoder, and include it in the generation process in rather **dense way***

- + What do you mean by **dense way**?!
 - Simply speaking, the condition embedding should **show up every layer**
- + How can we do it?!
 - We can simply have it **in every layer**; there are **other approaches as well**

Practical Implementation: Cross Attention

One popular approach is to use *cross attention*



Practical Implementation: FiLM

Another approach is to use *feature-wise linear modulation (FiLM)*: say we have a *particular layer* $L_{\mathbf{w}}$ in our basic model

it computes features $y_{\ell} = L_{\mathbf{w}}(y_{\ell-1})$ from outputs of previous layer $y_{\ell-1}$

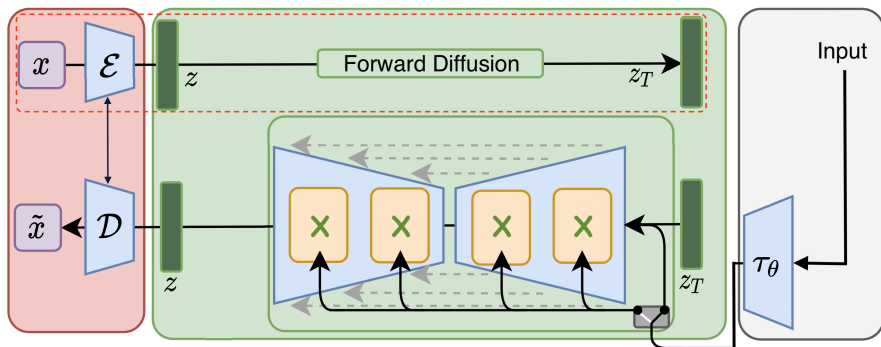
FiLM applies *conditioning* as follows

- ① it computes *embeddings* $\{a, b\} = \text{embd}_{\theta}(u)$
- ② it *conditions* all layers in the model as

$$\text{FiLM}\{L_{\mathbf{w}}|u\} = a \circ L_{\mathbf{w}} + b$$

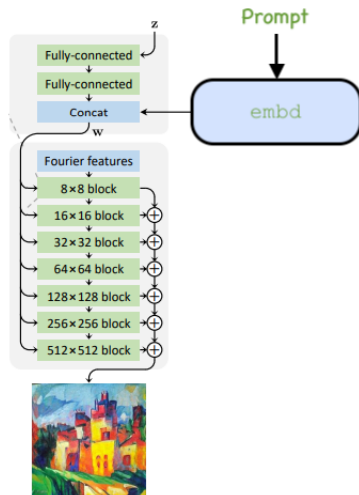
Stable Diffusion: Cross Attention

A well-known example of conditioning by *cross attention* is *Stable Diffusion*



Conditional StyleGAN: *FiLM*

*Conditional StyleGAN was one of famous models using **FiLM***



Final Notes

All we need for *multimodality* is *conditioning*

- *Conditioning* is implemented by *embedding* the prompt
- To keep the *conditioning* strong, the embedding should integrate *densely*

We can make all generative models: you could take a look at

- Conditional AR Models
- Conditional EBMs
- Conditional GANs
- Conditional VAEs
- Conditional Diffusions

You Know a Lot . . .

It was a great pleasure to learn generative models together in this semester!

! Important

*Any **knowledge** comes hand-in-hand with **responsibility**!*

You know a lot about **Generative AI**: this means that

- You should be **self-confident** to use it for problem solving
- You are **responsible** to **inform people** about what it can do
 - ↳ An LLM can learn the writing style of an author from its books
 - ↳ The author has the right to know this before sharing their books!
- You should **educate yourself** about privacy and ethical aspects
 - ↳ We are still on the beginning and many things have not been regulated!